

PELLEY BAY, NU, Canada

WMO: 719180

Lat: 68.437N

Lon: 89.726W

Elev: 1064

StdP: 14.14

Time Zone: -7.00 (NAM)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-43.5	-40.9	-41.3	0.5	-34.4	-40.8	0.5	-34.2	44.3	-14.3	40.8	-20.1	15.2	270	1.196

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.9	63.2	54.8	59.2	51.8	55.5	48.9	54.7	61.4	50.3	56.4	47.4	52.6	13.0	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
50.5	56.6	57.2	46.2	48.1	53.2	43.5	43.4	49.6	23.4	61.5	20.8	56.8	19.1	52.5	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 39.5	35.4	30.9	DB	-45.3	68.6	5.1	5.0	-48.9	72.2	-51.9	75.1	-54.7	77.9	-58.4	81.5	(4)
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	7.2	-24.7	-25.0	-17.8	-2.9	14.5	34.0	46.2	41.4	28.8	13.0	-5.9	-16.9	(6)	
	DBStd	26.84	10.25	10.56	12.30	10.96	8.47	8.43	7.20	7.27	6.86	9.42	11.98	11.71	(7)	
	HDD50	15680	2316	2101	2103	1588	1099	489	166	285	637	1147	1676	2073	(8)	
	HDD65	21080	2781	2521	2568	2038	1564	931	583	731	1087	1612	2126	2538	(9)	
	CDD50	74	0	0	0	0	0	8	47	19	0	0	0	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	14.8	15.6	14.2	13.5	14.7	16.8	15.0	13.8	14.4	14.4	16.1	14.7	14.6	(14)	
	PrecAvg	9.4	0.1	0.1	0.2	0.4	0.8	0.8	1.5	2.1	1.6	1.1	0.4	0.2	(15)	
(16) Precipitation	PrecMax	17.2	0.7	0.7	1.3	1.9	2.7	2.1	4.6	5.6	3.9	3.9	1.3	1.7	(16)	
	PrecMin	4.4	0.0	0.0	0.0	0.0	0.0	0.0	0.5	0.3	0.3	0.2	0.0	0.0	(17)	
	PrecStd	3.0	0.2	0.2	0.3	0.4	0.6	0.5	0.8	1.3	0.9	0.7	0.3	0.3	(18)	
	PrecStd	3.0	0.2	0.2	0.3	0.4	0.6	0.5	0.8	1.3	0.9	0.7	0.3	0.3	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.9	5.7	15.9	26.3	33.2	65.0	69.9	67.5	52.1	33.6	24.2	19.9	(19)	
		MCWB	11.9	4.8	15.3	25.7	30.8	50.4	56.5	58.9	48.7	33.2	23.8	17.8	(20)	
	2%	DB	0.5	1.1	10.0	21.0	30.7	57.4	64.3	61.1	46.6	29.3	19.7	11.4	(21)	
		MCWB	0.0	0.1	9.2	20.7	29.9	48.5	56.0	55.6	42.3	28.8	19.4	10.7	(22)	
	5%	DB	-5.1	-3.9	5.4	17.2	28.4	51.4	60.8	56.4	42.8	26.9	15.7	5.9	(23)	
		MCWB	-5.5	-4.4	4.8	16.6	27.8	45.4	52.5	50.5	40.7	26.5	15.3	5.4	(24)	
	10%	DB	-9.6	-8.6	0.5	13.5	25.8	46.0	57.8	52.4	38.5	24.6	12.0	0.1	(25)	
		MCWB	-9.9	-9.2	-0.2	12.8	25.3	41.7	50.1	46.5	37.5	24.0	11.5	-0.3	(26)	
(28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.4	3.9	10.2	23.8	31.0	51.3	59.0	60.3	49.5	34.7	22.0	13.6	(27)	
		MCDB	8.4	4.4	10.8	24.2	31.3	56.9	65.2	66.0	51.7	35.1	22.2	13.9	(28)	
	2%	WB	-1.2	-0.5	3.9	17.5	28.7	47.7	55.1	55.7	44.2	28.4	16.1	9.7	(29)	
		MCDB	-0.7	0.0	4.3	18.0	29.4	54.8	63.6	60.7	47.1	28.6	16.3	10.5	(30)	
	5%	WB	-7.9	-3.4	-1.3	13.8	26.4	44.8	51.8	47.5	40.1	24.7	11.3	3.9	(31)	
		MCDB	-7.5	-2.8	-0.9	14.6	26.8	49.4	59.1	51.5	42.4	25.0	11.7	4.5	(32)	
	10%	WB	-13.4	-7.9	-4.5	9.0	23.7	41.9	48.8	44.2	36.3	22.0	4.0	-4.0	(33)	
		MCDB	-13.1	-7.5	-4.0	9.6	24.2	45.6	55.0	48.0	37.4	22.5	4.5	-3.7	(34)	
(36) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.0	10.5	10.5	8.3	7.9	10.9	9.2	6.3	7.1	8.8	8.3	(35)	
		MCDBR	12.5	11.5	12.3	11.5	8.9	14.5	14.5	14.8	11.5	7.9	11.6	11.7	(36)	
		MCWBR	16.7	11.2	11.0	9.9	10.2	7.6	8.6	12.3	8.9	9.6	13.7	9.7	(37)	
	5% WB	MCDBR	17.0	11.3	11.3	10.2	10.1	10.0	12.6	14.9	10.4	8.6	14.4	9.4	(38)	
		MCWBR	16.8	11.2	11.2	10.4	10.1	7.8	9.3	11.8	8.9	8.6	13.9	9.4	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.142	0.197	0.243	0.285	0.310	0.308	0.339	0.310	0.282	0.227	0.147	N/A	(40)	
	taud		1.893	2.097	2.201	2.136	2.144	2.375	2.415	2.518	2.471	2.104	1.908	N/A	(41)	
	Ebn at Noon		43	201	256	272	277	283	269	267	245	174	33	0	(42)	
	Edh at Noon		2	14	23	34	39	32	30	24	19	13	2	0	(43)	
(44) All-Sky Solar Radiation	RadAvg		9	143	639	1404	1944	1896	1625	1098	536	167	27	0	(44)	
	RadStd		1	10	23	47	114	167	160	85	46	18	2	0	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (4 neighbors)	+1.64	+1.71	N/A	N/A	N/A	N/A	-785	-778	N/A	N/A

Nomenclature: See separate page